

Andes Virus Outbreak

MV Hondius Cruise Ship · 2026

26 May 2026

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Outbreak summary · 26 May 2026

11 cases in analysis (9 confirmed · 2 probable) · 3 deaths · 06 Apr – 14 May 2026

Background

Andes virus (ANDV) is the only hantavirus documented to transmit person-to-person, in addition to the usual rodent-to-human route. The 2026 outbreak is linked to the **MV Hondius** expedition cruise ship. Cases first appeared in early April 2026; by the time infection was confirmed, passengers and crew had dispersed across Europe, Asia, and the Americas. Whole-genome sequencing confirms ANDV as the causative agent.

Methods

Transmission trees were inferred with **outbreaker2**, integrating symptom onset dates and genome sequences. Serial interval was modelled as Gamma(mean = 18 d, CV = 1/3). MCMC: 100,000 iterations, thinned every 50 steps, 10% burn-in. L-segment sequences were retrieved from [Pathoplexus](#) for the 4 of 11 cases with deposited accessions. The S and M segments are identical across all cases; the L segment carries the only two SNPs observed in the outbreak and is therefore the sole source of genetic signal. Remaining cases contribute temporal data only. Because near-zero genetic variation renders the per-site mutation rate (μ) non-identifiable from the data, μ was fixed at 1.7×10^{-5}

substitutions/site/generation, derived from the published ANDV substitution rate of $\sim 3.5 \times 10^{-4}$ subs/site/year and an 18-day generation interval.

Results

Cases

Table 1: Confirmed and probable cases.

Case ID	Status	Role	Age (yr)	Sex	Nationality	Onset date	Outcome	Sequence
1	Probable	Passenger	70	Male	Dutch	06 Apr 2026	Fatal	—
2	Confirmed	Passenger	69	Female	Dutch	24 Apr 2026	Fatal	PP_006WDKH.1
3	Confirmed	Passenger	NA	Male	British	24 Apr 2026	—	PP_00WANE.2
8	Confirmed	Crew	41	NA	Dutch	27 Apr 2026	—	PP_006W6RC.2
4	Confirmed	Passenger	NA	Female	German	28 Apr 2026	Fatal	—
12	Probable	Passenger	NA	Male	British	28 Apr 2026	—	—
7	Confirmed	Crew	56	Male	British	30 Apr 2026	—	PP_006W3U9.1
5	Confirmed	Passenger	NA	Male	Swiss	01 May 2026	—	PP_006WBLH.1
15	Confirmed	Passenger	NA	Female	French	10 May 2026	—	—
18	Confirmed	Passenger	NA	NA	Spanish	10 May 2026	—	—
20	Confirmed	Passenger	70	NA	Canadian	14 May 2026	—	—

Genomic Analysis

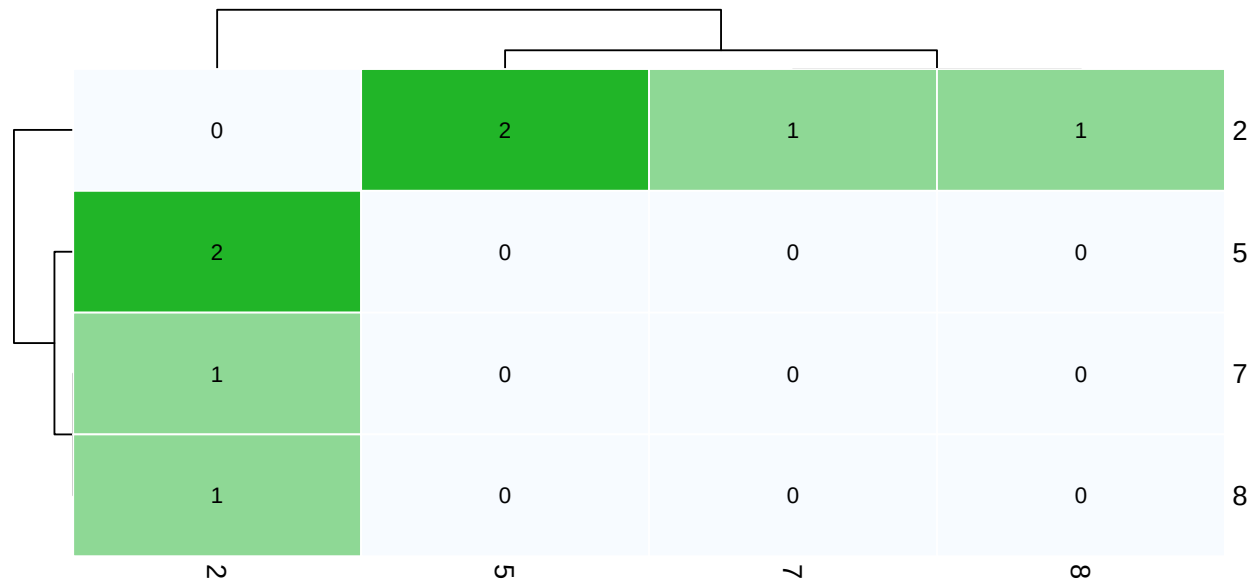


Figure 1: Pairwise SNP distance matrix for sequenced cases (L segment). Values are raw nucleotide differences. Near-zero distances confirm a near-clonal cluster consistent with a single spillover event.

Table 2: Convergence diagnostics across 4 independent chains. $R\text{-hat} < 1.01$ and $ESS > 400$ indicate good convergence and mixing.

Parameter	R-hat	97.5% CI	ESS	R-hat: Green < 1.01 · Yellow 1.01–1.05 · Red 1.05+ samples (4 chains $\times$ 1800 samples per chain).
Reporting rate ()	1.000	1.000	7,200	

Model Diagnostics

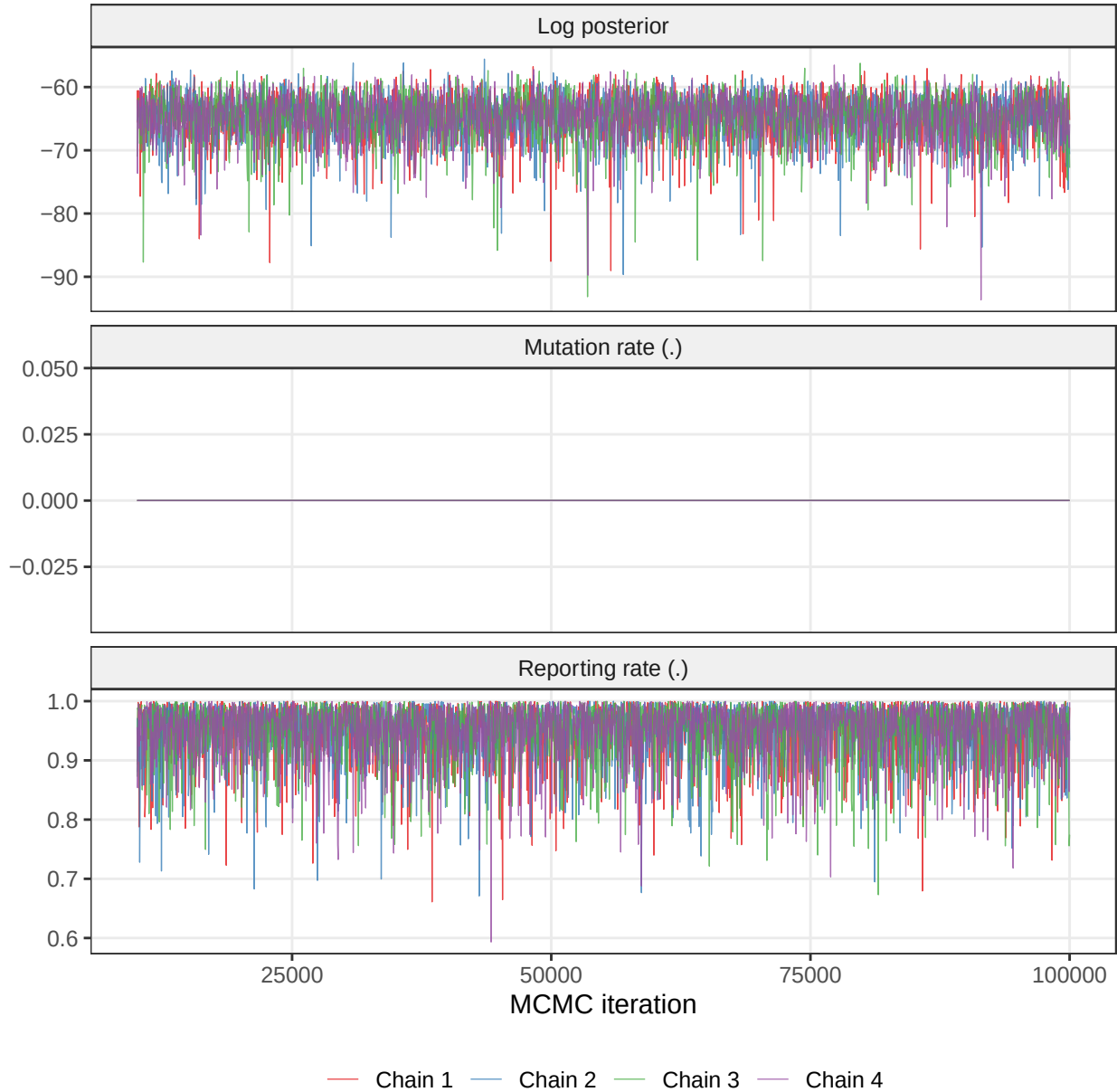


Figure 2: MCMC trace plots for scalar parameters (post-burnin). Stationary, well-mixed traces indicate convergence.

Transmission Tree

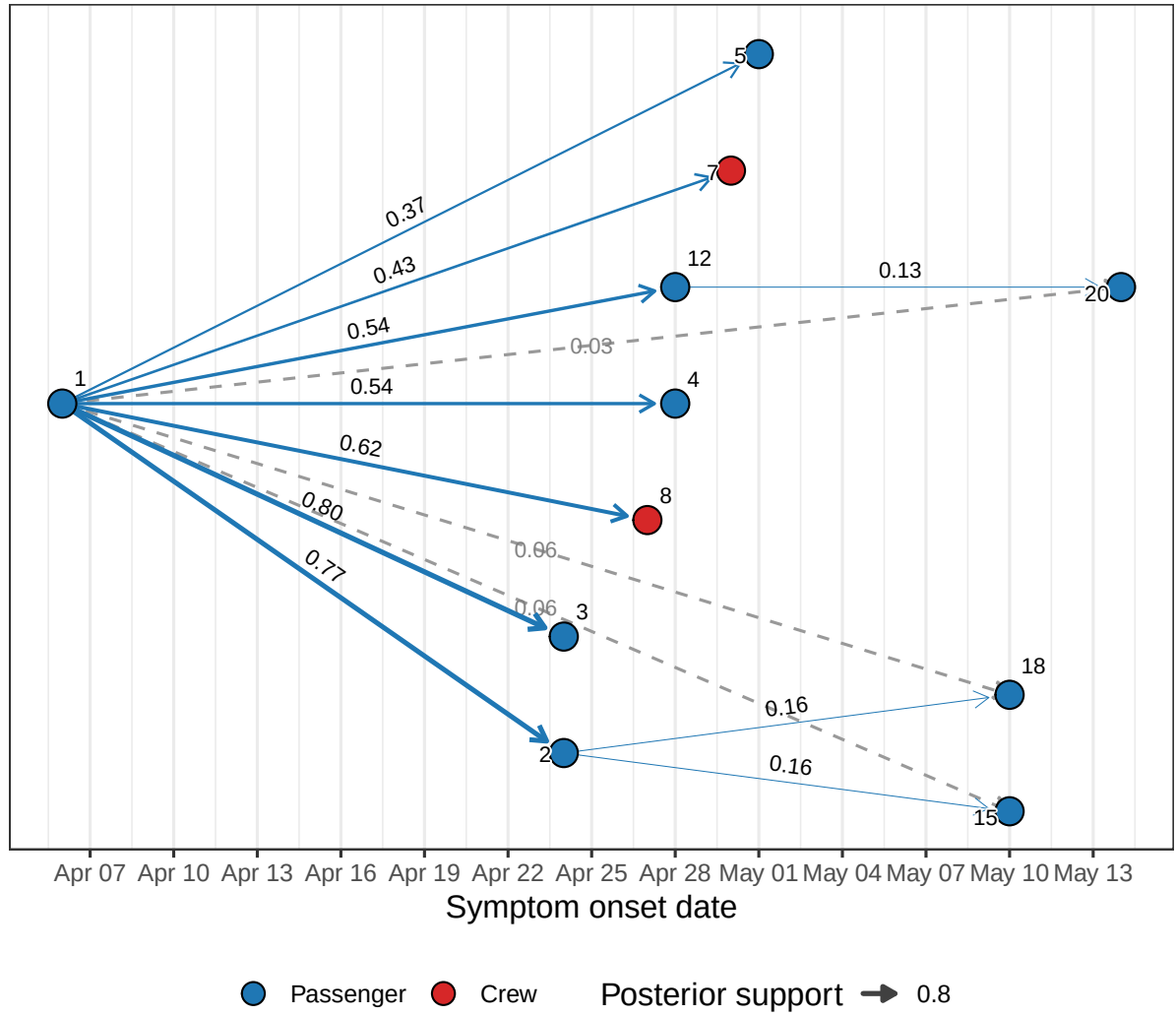


Figure 3: Consensus transmission tree (coloured edges), where each case is assigned its most frequently inferred ancestor across the posterior sample. Grey dashed arrows show the posterior support for direct transmission from case 1 to all cases not already linked to it in the consensus tree, consistent with a superspreading event.

Ancestry Uncertainty

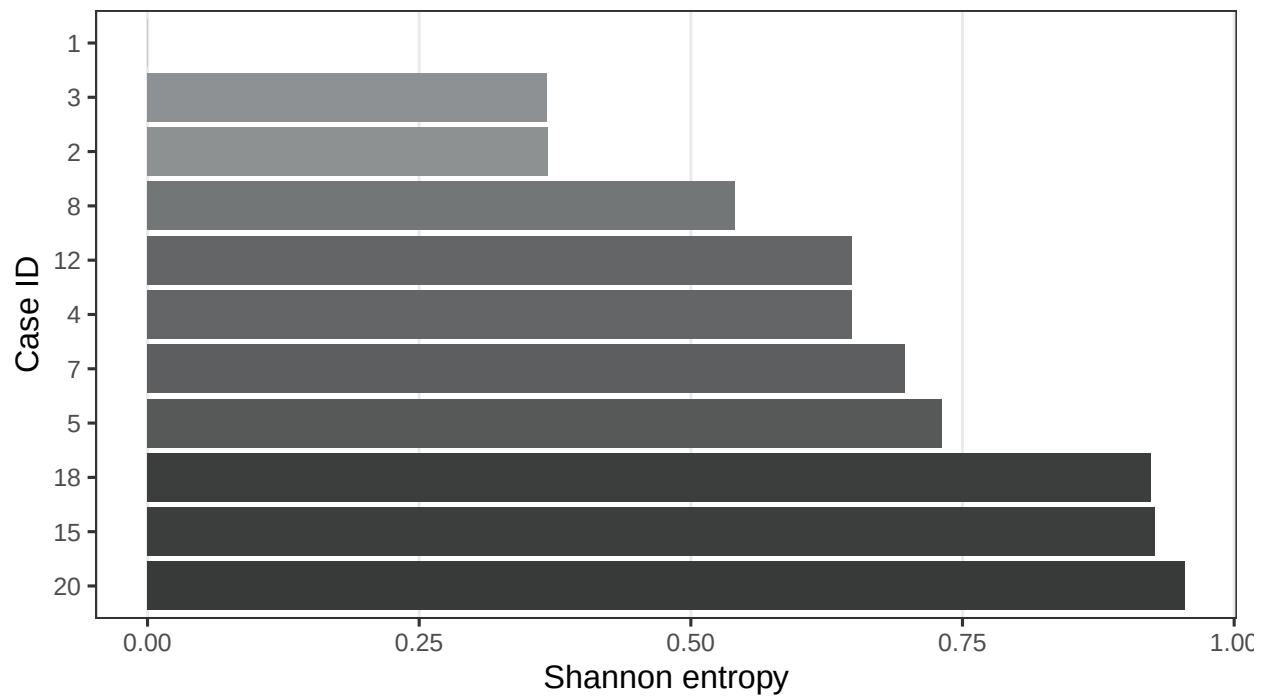


Figure 4: Shannon entropy of each case's inferred source. Entropy = 0 means a single unambiguous infector.

Dates of Infection

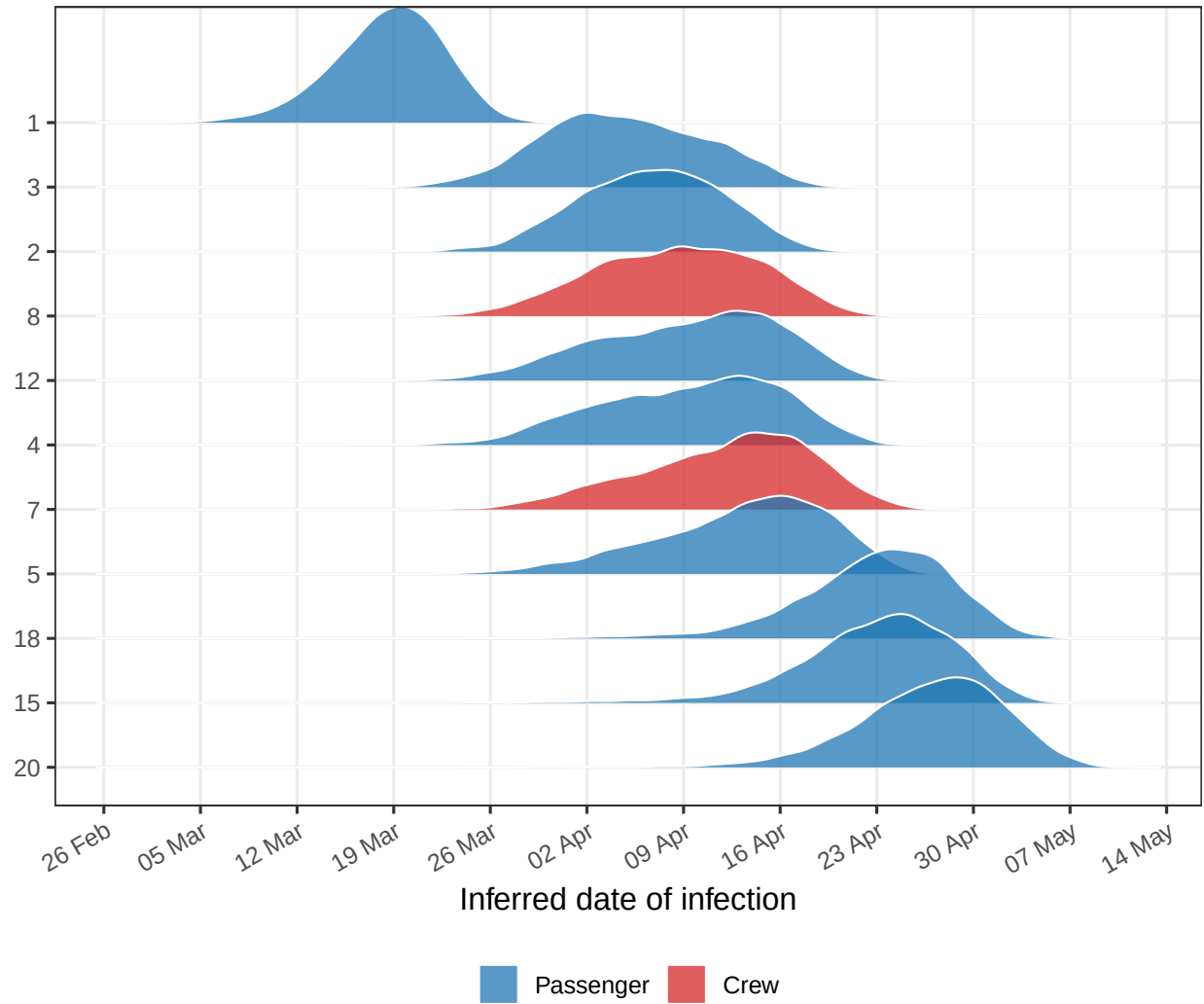


Figure 5: Posterior distribution of inferred infection dates per case. Width reflects uncertainty; cases ordered by mean inferred infection date.

Imports and Unobserved Transmission

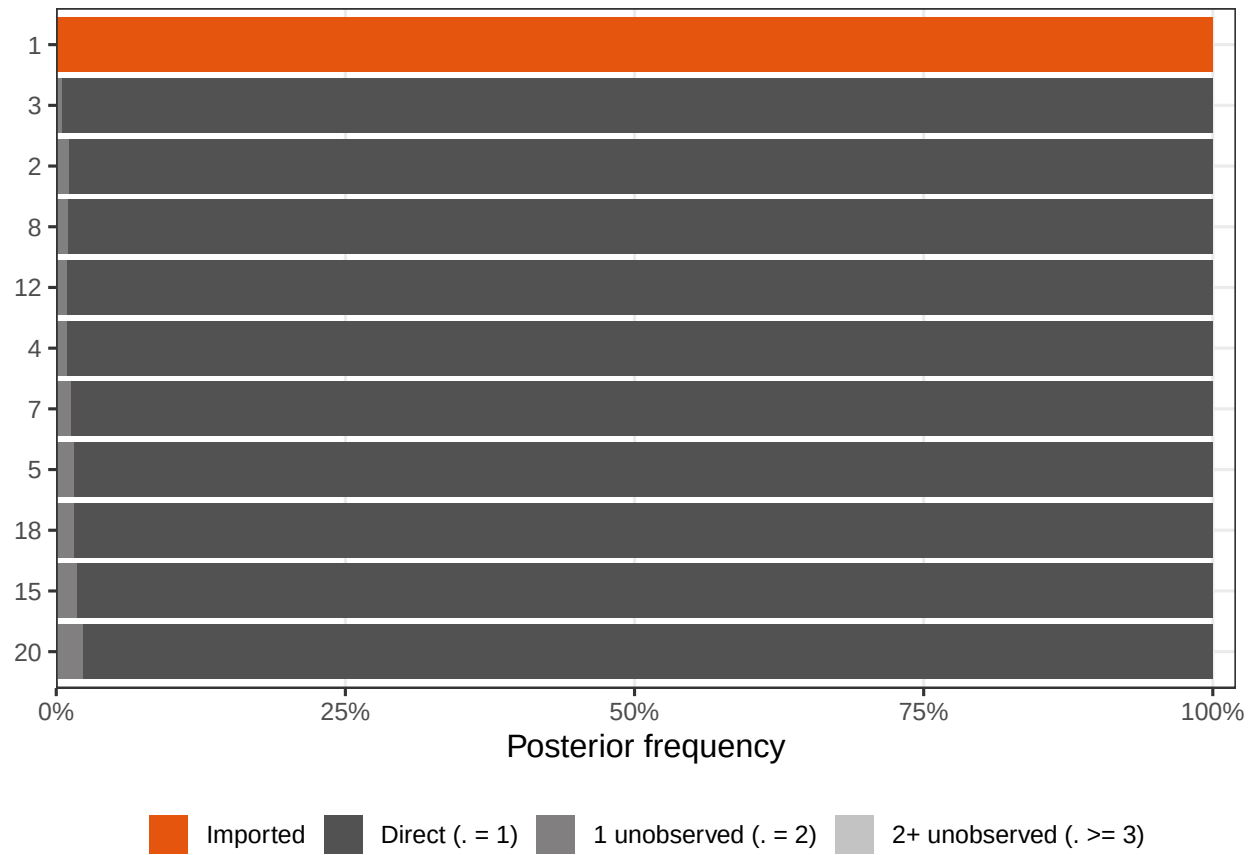


Figure 6: Posterior probability of each case being imported, directly linked, or connected via unobserved intermediate cases ().

Reproduction Number

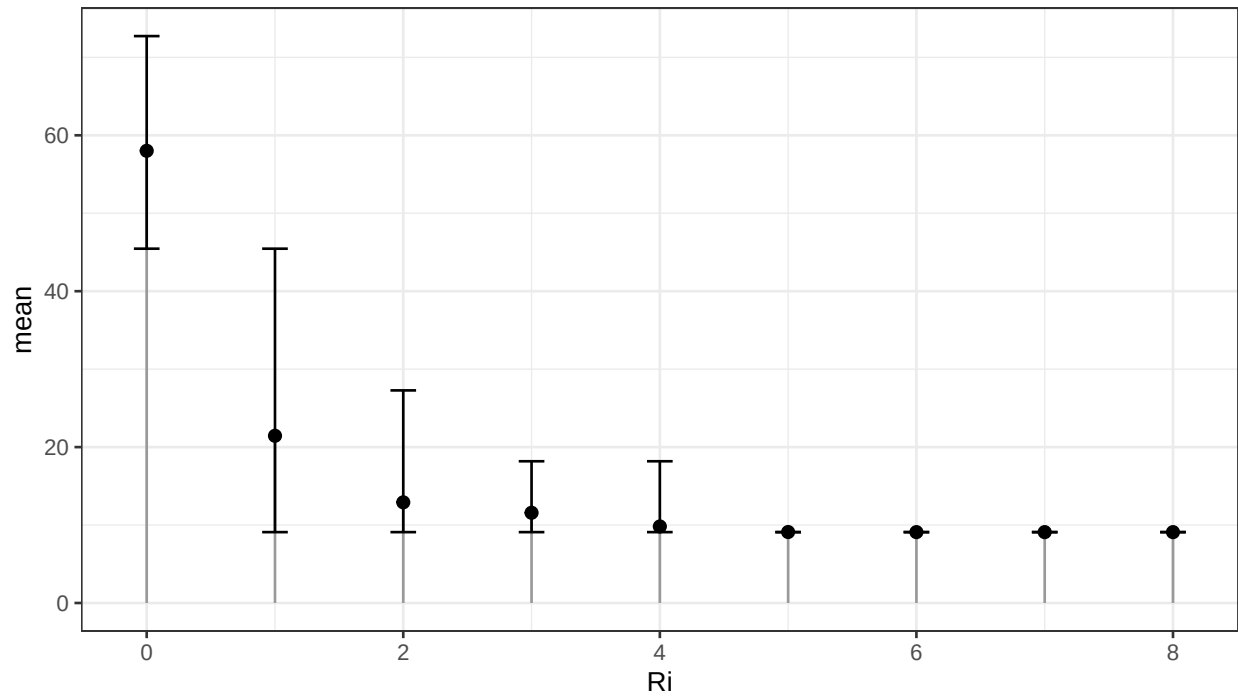


Figure 7: Posterior distribution of the offspring distribution.

Data

Data available under [CC BY 4.0](#) at [kraemer-lab/Hondius_hantavirus_h2026](#). Sequences retrieved from [Pathoplexus](#). Report source: `report.qmd`.